CONNECTORS, COAXIAL, RADIO FREQUENCY, SERIES TPS

1. SCOPE

1.1 This specification covers the following types of weatherproof connectors of the "TPS" series, designed to produce minimum electrical discontinuities in a solid dielectric coaxial cable of characteristic impedance of 50 ohms, up to a frequency of 10,000 megacycles and rated at 1,500 volts r.m.s. at sea level. (See 6.4.)

Connector, Receptacle, Electrical	UG-1364( )/U
Connector, Receptacle, Electrical	UG-1365( )/U
Connector, Plug, Electrical	UG-1366( )/U
Adapter, Connector	UG-1367( )/U
Adapter, Connector	UG-1368( )/U
Connector, Plug, Electrical	UG-1412( )/U
Connector, Receptacle, Electrical	UG-1413( )/U
Connector, Plug, Electrical	UG-1415( )/U
Connector, Plug, Electrical	UG-1416( )/U
Adapter, Connector	UG-1426( )/U
Adapter, Connector	UG-1427( )/U
Adapter, Connector	UG-1428( )/U
Adapter, Connector	UG-1429( )/U
Connector, Receptacle, Electrical	UG-1443( )/U
Connector, Receptacle, Electrical	UG-1471( )/U

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-389	Plastics, Molding, FEP-Fluorocarbon Resin.
QQ-B-613	Brass, Leaded and Non-Leaded, Plate, Rolled Bar, Sheet and Strip.

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SPECIFICATIONS (Cont'd)

QQ-B-626	Brass, Leaded and Non-Leaded, Rods, Shapes, Forgings and Flat Products with Finished Edges, Bars, Flat Wire and Strips.
QQ-C-530	Copper-Beryllium Alloy Bars, Rods, and Wire.
QQ-C-533	Copper-Beryllium Alloy Strip.
QQ-S-365	Silver Plating, Electrodeposited, General Requirements for.
QQ-S-571	Solder, Lead Alloy, Tin-Lead Alloy, Flux Cored, Ribbon and Wire.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Fiber.
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof, For Packaging and Sealing.
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-4229	Boxes; Paperboard, Metal-Stayed.
MIL-R-5847	Rubber Silicone, Low and High Temperature and Tear Resistant.
MIL-H-7199	Heat Treatment of Copper-Beryllium Alloys, Process for.
MIL-B-10377	Boxes, Wood, Cleated, Veneer, Paper-Overlaid.
MIL-L-10547	Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-M-14077	Molding Plastic, Polytetrafluoroethylene.
MIL-G-45204	Gold Plating (Electrodeposited).

STANDARDS

FEDERAL

No. 406	Plastics: Method of Testing.

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STANDARDS (Cont'd)

MILITARY

MS33586	Metals, Definition of Dissimilar.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking for U. S. Military Equipment.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

ELECTRONICS COMMAND

SC-D-358720	Dimensional Interchangeability Requirements for Male Connectors.
SC-D-358721	Dimensional Interchangeability Requirements for Female Connectors.
SC-D-358722	Outline and Mounting Dimensions for Receptacles, Jacks and Adapters.
SC-DL-358723	Connector, Receptacle, Electrical UG-1364( )/U (Bulkhead Jack for RG-58 Cable).
SC-DL-358724	Connector, Receptacle, Electrical UG-1365( )/U
SC-DL-358725	Connector, Plug, Electrical UG-1366( )/U (For RG-58 Cable).
SC-DL-358726	Adapter, Connector UG-1367( )/U (Bulkhead, Female-Female, TPS to TPS).
SC-DL-358727	Adapter, Connector UG-1368( )/U (Bulkhead, Female-Female, TPS to BNC).
SC-DL-358728	Connector, Plug, Electrical UG-1412( )/U (For RG-223 Cable)
SC-DL-358729	Connector, Receptacle, Electrical UG-1413( )/U (Bulkhead Jack for RG-223 Cable).
SC-DL-358730	Connector, Plug, Electrical UG-1415( )/U. (Jack for RG-58 Cable).
SC-DL-358731	Connector, Plug, Electrical UG-1416( )/U (Jack for RG-223 Cable).

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DRAWINGS (Cont'd)

SC-DL-358732	Adapter, Connector UG-1426( )/U (TPS-Female to N-Female).
SC-DL-358733	Adapter, Connector UG-1427( )/U (TPS-Female to N-Male).
SC-DL-358734	Adapter, Connector UG-1428( )/U (TPS-Male to N-Female).
SC-DL-358735	Adapter, Connector UG-1429( )/U (TPS-Male to N-Male).
SC-DL-358736	Connector, Receptacle, Electrical UG-1471( )/U.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois.)

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

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3. REQUIREMENTS

3.1 General.- The connectors covered by this specification shall be constructed using the drawings specified on the drawing and data lists referenced in 2.1 as a guide for informational purposes. Dimensional deviations from these drawings, except for the interchangeability, and outline and mounting dimensions specified in 3.10 and 3.11 respectively, shall be permitted. Compliance with the performance requirements of this specification, regardless of design details employed, shall determine the acceptability of these connectors.

3.2 Preproduction samples.- The contractor shall furnish six preproduction samples of each type of connector for approval, provided the Invitation for Bids and the Contract requires samples. (See 4.3.)

3.3 Materials.- Unless otherwise specified, the material for each part shall be as specified herein. When a definite material is not specified, a material shall be used which will enable the connectors to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.3.1 Brass.- Brass shall conform to composition 11, half-hard, of Specification QQ-B-613, or composition 22, half-hard, of Specification QQ-B-626, as applicable.

3.3.2 Beryllium copper.- Beryllium copper shall conform to Condition H of Specification QQ-C-530 or QQ-C-533, as applicable. Parts fabricated of copper beryllium shall be heat treated to Condition HT in accordance with Specification MIL-H-7199, after machining and forming.

3.3.3 Insulator.- The insulator shall be made of polytetrafluoroethylene (TFE) in accordance with Class 1 of Specification MIL-G-14077 or fluorinated ethylene propylene (FEP) in accordance with Specification L-P-389. The insulator shall have no visible flaws or cracks, and shall have a dielectric constant and dissipation factor (tangent of loss angle) not greater than 2.1 and 0.0002, respectively, when measured between 50 and 100 megacycles in accordance with Method 4021 of Federal Standard No. 406. Only clean, unreprocessed, virgin TFE or FEP shall be used.

3.3.4 Gasket.- The gasket shall be made of silicone rubber conforming to Class II, Grade 50 or 60 of Specification MIL-R-5847 except that the oil-immersion test is not applicable.

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3.3.5 Solder. - Solder shall conform to composition Sb5 of Specification QQ-S-571.

3.3.6 Dissimilar metals. - Unless suitably protected against electrolytic corrosion, dissimilar metals, as defined in Standard MS33586, shall not be used in intimate contact. Where it is necessary that any combination of such dissimilar metals be assembled, an interposing metal compatible to each shall be used.

3.3.7 Corrosion resistant materials. - Materials shall be of the corrosion resistant types or shall be processed to resist the corrosion tests specified herein, after which the connectors shall be serviceable.

3.4 Design and construction. - The design, construction, and physical dimensions of the mating ends of the connectors shall be as specified on Drawings SC-D-358720 and SC-D-358721, as applicable.

3.4.1 Silver plating. - The silver plating on metal parts shall be not less than 0.0005 inch thick for exposed surfaces and 0.0002 inch thick for internal surfaces. An exposed surface shall be defined as a surface which is subjected directly to the environments in the mated condition (i.e. outer shell, coupling nut, etc.) Silver plating shall be in accordance with Specification QQ-S-365. Dimensions of metal parts shall include the plating.

3.4.2 Gold plating. - All surfaces of gold plated metal parts shall have a silver plating not less than 0.0002 inch thick and a gold plating in accordance with Specification MIL-G-45204, not less than 0.00005 inch over the silver plating.

3.4.3 Screw-threads. - Except for non-replaceable structural parts, screw threads shall be in accordance with Handbook H28 and shall be NEF-2 fit after plating.

3.4.4 Contacts. - The terminal ends of the contacts shall be hot tin-dipped or gold plated to facilitate soldering.

3.4.4.1 Male (pin) contacts. - Male contacts shall be fabricated of half-hard brass material specified in 3.3.1.

3.4.4.2 Female contacts. - The female center and outer contacts shall be fabricated of beryllium copper material specified in 3.3.2.

3.5 Coupling sleeve rotation. - The torque required to effect rotation of the coupling sleeve relative to the connector body shall be not greater than 0.75 inch-pound, when tested as specified in 4.7.6.

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3.6 Cable retention. - The cable properly attached to the connector shall withstand a minimum pull force of 45 pounds without breaking or movement from the initial fixed position, when tested as specified in 4.7.7.

3.7 Form over. - Any form over in the connectors shall withstand the force specified without any evidence of damage to the form over, when tested as specified in 4.7.8.

3.8 Cable clamping. - The method used for securing the cable to the connector shall not require the use of special tools, shall be simple, and shall require no combing of the braid to meet the cable retention requirement of 3.6.

3.9 Assembly instructions. - Instructions for the assembly of cables to connectors shall be concise and shall be included in each package with the connector, as applicable.

3.10 Interchangeability dimensions. - Like units and assemblies shall be physically and functionally interchangeable. Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc. The interchangeability dimensions for the male and female connectors shall be as specified on Drawings SC-D-358720 and SC-D-358721.

3.11 Outline and mounting dimensions. - The outline and mounting dimensions of the receptacles, jacks, and adapters, shall be as specified on Drawing SC-D-358722.

3.12 Marking. - Connectors shall be marked in accordance with Standard MIL-STD-130 with only the type designation and the manufacturer's code (i.e. UG-1366/U 95172). Marking shall be in depressed characters, 3/64 inch high, in a convenient place.

3.13 Dielectric withstanding voltage. - There shall be no evidence of voltage breakdown between the center conductor properly positioned and the outer shell of the connectors, and between the center conductor and the braid of the cable for the adapters, when tested as specified in 4.7.3.

3.14 Insulation resistance. - The insulation resistance between the center conductor and outer shell of mated connectors shall be a minimum of 100,000 megohms initially and after moisture resistance, when tested as specified in 4.7.4 and 4.7.14.

3.15 Contact resistance. - The resistance measurements on all connectors (excluding adapters) of mated inner contacts and of mated outer contacts measured individually shall not exceed 0.0025 and 0.0003 ohms, respectively, and the contact resistance limits of the adapters shall be 0.005 and 0.0006 ohms, respectively, when tested as specified in 4.7.5.

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3.16 Voltage standing wave ratio (VSWR). - The voltage standing wave ratio of the mated connectors shall not exceed a value of 1.35: 1 up to 10,000 megacycles, when tested as specified in 4.7.9.

3.17 Leakage. - The insulation resistance between the center conductor and the outer shell of the mated connectors shall exceed 100,000 megohms initially and after the leakage test, when tested as specified in 4.7.4 and 4.7.10.

3.18 Durability. - When tested as specified in 4.7.11 for 500 applications of matings and unmatings, the connectors shall meet the requirements of 3.13.

3.19 Temperature cycling. - The connectors shall exhibit no evidence of cracking, streaking, corrosion, variation in dimensions, changes in operating and electrical characteristics, or any other damage, when tested as specified in 4.7.12.

3.20 Corrosion. - The connectors shall exhibit no evidence of destructive corrosion or pitting, when tested as specified in 4.7.13. Destructive corrosion shall be defined as follows:

3.20.1 Mechanical. - After exposure to the salt spray test (see 4.7.13), the coupling sleeve shall meet the requirements of 3.5.

3.20.2 Electrical. - After exposure to the salt spray test (see 4.7.13), the contact resistance of the inner and outer contacts of the connectors shall be a maximum of 0.004 and 0.0005 ohms respectively, and the contact resistance limits of the adapters shall be 0.008 and 0.001 ohms respectively, when measured as specified in 4.7.5.

3.21 Moisture resistance. - When tested as specified in 4.7.14, the connectors shall meet the requirements of 3.13 and 3.14.

3.22 Vibration and shock. - When tested as specified in 4.7.15 and 4.7.16, there shall be no evidence of cracking, breaking, or loosening of any of the component parts of the connectors and there shall be no evidence of interruption of electrical contact of the mated connectors.

3.23 Workmanship. - Workmanship shall be such as to permit the connectors to meet all the requirements of this specification. The connectors shall be processed in such a manner as to be uniform in quality, and shall be free from sharp edges and burrs, except where sharp edges are required for mechanical or electrical reasons. All solder joints shall be thoroughly cleaned.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of connectors before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.6)

4.3 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents, covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (see tables II, III, and IV, respectively).

Table I - Preproduction inspection

Inspection	Req Para	Insp Para
Vibration	3.22	4.7.15
Shock	3.22	4.7.16
Temperature cycling	3.3.7 and 3.19	4.7.12
Salt spray	3.3.7 and 3.20	4.7.13
Durability	3.18	4.7.11
Leakage	3.17	4.7.10
Moisture resistance	3.21	4.7.14

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of the connectors before preparation for delivery:

Item	Where required
Brass	3.3.1
Copper beryllium	3.3.2
TFE or FEP	3.3.3
Silicone rubber	3.3.4
Solder	3.3.5
Dissimilar metals	3.3.6
Silver plating	3.4.1
Gold plating	3.4.2
Screw threads	3.4.3
Marking	3.12

4.5 Quality conformance inspection of connectors before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

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Table II - Group A inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and mechanical				
Materials	3.3	4.7.1	1% for the group	4% for the group
Design and construction	3.4, 3.4.1, 3.4.2 3.4.3 and 3.4.4	4.7.1		
Cable clamping	3.8	4.7.1		
Assembly instructions	3.9	4.7.1		
Outline and mounting dimensions	3.11	4.7.1		
Marking	3.12	4.7.1		
Workmanship	3.23	4.7.1		
Interchangeability dimensions	3.10	4.7.2	1%	4%
Electrical				
Dielectric withstanding voltage	3.13	4.7.3	1% for the group	
Insulation resistance	3.14	4.7.4		
Contact resistance	3.15	4.7.5		

* All electrical defects are considered major

4.5.2 Group B inspection.- This inspection, including sampling shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected and met the group A inspection.

4.5.2.1 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table III - Group B inspection

Inspection	Req Para	Insp Para	AQL
Coupling sleeve rotation	3.5	4.7.4	6.5%
Form over	3.7	4.7.8	6.5%

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4.5.3 Group C inspection.- This inspection shall be as listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV.- Group C inspection

Inspection	Req Para	Insp Para
Cable retention	3.6	4.7.7
Voltage standing wave ratio	3.16	4.7.9
Durability	3.18	4.7.11

4.5.3.1 Sampling for inspection of connectors.- Two of each type of connector for group C inspection shall be selected each month, without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.3.)

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.7 Methods of examination and test.- When mated pairs of connectors are specified, the test sample shall consist of the connector under test with an appropriate mating connector. Unless otherwise specified, all tests for adapters and receptacles shall be performed, mated with their associated mating connectors.

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4.7.1 Visual and mechanical inspection.- Connectors shall be inspected to verify that the materials, construction, cable clamping, assembly instructions, outline and mounting dimensions, marking, and workmanship are in accordance with the applicable requirements. (See 3.3, 3.4, 3.8, 3.9, 3.11, 3.12 and 3.23)

4.7.2 Inspection for interchangeability dimensions.- The interchangeability dimensions of the male and female connectors shall be gaged at the specific points indicated on Drawing SC-D-358720 and SC-D-358721, to determine compliance with the interchangeability requirement of 3.10. When a listed dimension is not within specified limits, it shall be considered a major defect.

4.7.3 Dielectric withstanding voltage.- Connectors mated and assembled to their appropriate cables, and adapters mated to a plug or jack, as applicable, assembled to its appropriate cable shall be tested in accordance with Method 301 of Standard MIL-STD-202. The test voltage shall be raised from zero to 1500 volts rms at the rate of 100 volts per second and held at that voltage for one minute applied between the inner and outer conductors of the connector and between the center conductor and the braid of the cable for the adapters. (See 3.13).

4.7.4 Insulation resistance.- Connectors mated and assembled to their appropriate cables, and adapters mated to a plug, or jack, as applicable, assembled to its appropriate cable shall be tested in accordance with test condition A of Method 302 of Standard MIL-STD-202. (See 3.14) Application of voltage shall be between the center conductor and the outer shell of the mated connectors and between the center conductor and braid of the cable for the adapters.

4.7.5 Contact resistance.- Connectors shall be tested in accordance with Method 307 of Standard MIL-STD-202. The following details shall apply:

a. The electrical resistance of the mated connector contacts shall be calculated by measuring the potential drop across the mated contacts when carrying a current of 7.5 and 30 amperes in the inner and outer contacts, respectively. (See 3.15)

b. Before making the initial contact resistance measurement, the contacts shall be inserted and withdrawn five times.

c. The connections to the contacts shall be soldered.

d. The wire used for connecting the contacts to the driving circuit shall be of such physical size as to insure not more than a 30°C temperature rise when the specified current is flowing in the circuit.

4.7.6 Coupling sleeve rotation.- A torque shall be applied to the coupling sleeve about the axis of symmetry. The torque shall be measured with a 0 to 2 inch-pound torque wrench. (See 3.5)

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4.7.7 Cable retention.- Connectors assembled to their appropriate cables shall be subjected for one minute to an axial force of 45 pounds pulling the connector away from the cable. (See 3.6)

4.7.8 Form over.- A withdrawal force of 100 pounds shall be applied wherever a form over exists. The pull shall be made for one minute along the longitudinal axis of the two joined parts. (See 3.7)

4.7.9 Voltage standing wave ratio.- The voltage standing wave ratio of the connectors, mated and assembled to their appropriate cables, and adapters mated to a plug or jack, as applicable, assembled to its appropriate cable, with respect to a matched 50 ohm system, shall be measured as follows, at increments of 1000 megacycles, in the range of frequencies from 1000 to 10000 megacycles, as shown in the block diagram:

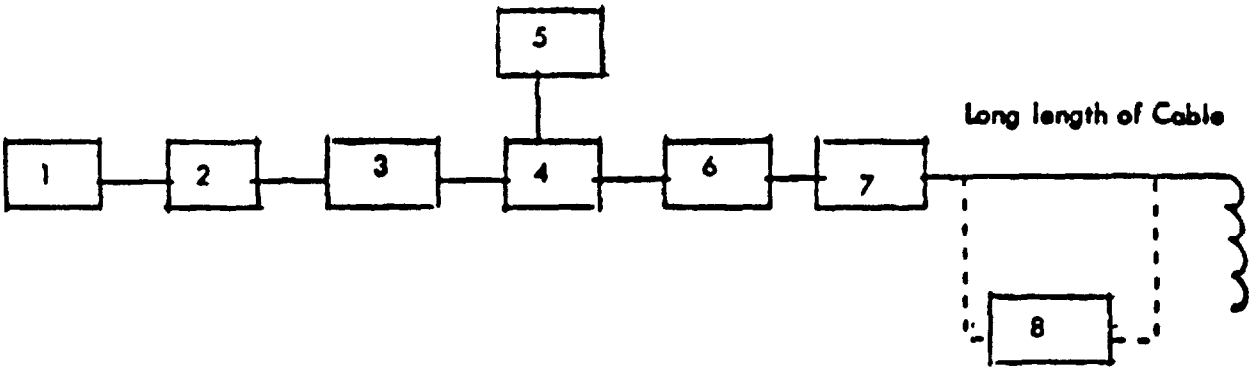
a. A long length of cable associated for connector use shall be assembled to the adapter. The length shall be adequate when shorting the far end of the cable does not shift the minimum of the standing wave. The connector assembly to be tested shall not be included in the initial calibration.

b. At each frequency that the VSWR is to be measured, the double stub tuner shall be set and locked so that the VSWR recorded on the standing wave indicator is 1.01 to 1 or better.

c. The cable shall be cut at a short distance from the adapter and a connector pair shall be inserted such that the insertion loss of the cable between the adapter and connector pair is negligible.

d. At each frequency, the proper pretuned stub tuner shall be inserted for that frequency. The VSWR reading obtained on the standing wave indicator shall be the cable to cable VSWR of the connector assembly.

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1. RF Signal Source
2. Frequency Meter
3. 10 db attenuator pad
4. Slotted Section
5. Standing wave Indicator
6. Double stub tuner
7. Transition to cable (when required)
8. Connector pair under test

4.7.10 Leakage.- Connectors mated and assembled to their appropriate cables shall be subjected to a gage pressure of 7.5 pounds per square inch for a period of one minute with the whole connector immersed in a tank of water at approximately 20°C. The free end of the cable shall be sealed to prevent water ingress. (See 3.17)

4.7.11 Durability.- Connectors assembled to their appropriate cables shall be subjected to 500 applications of insertions and withdrawals at a rate not to exceed 500 cycles per hour. The insertions and withdrawals shall be conducted in a manner similar to that which the connectors will be subjected to in service. (See 3.18)

4.7.12 Temperature cycling.- Connectors mated and assembled to their appropriate cables shall be subjected to five temperature cycles in accordance with Method 102A of Standard MIL-STD-202 as follows:

Polytetrafluoroethylene cables	Polyethylene cables	Time duration
-55°C	-55°C	30 minutes
+25°C	+25°C	15 minutes
+200°C	+85°C	30 minutes
+25°C	+25°C	15 minutes

Adapters and receptacles shall be subjected to the temperature cycle of polytetrafluoroethylene cables. (See 3.19)

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4.7.13 Salt spray.- Unmated connectors and connector contacts shall be tested in accordance with Method 101, Test Condition B, of Standard MIL-STD-202. At the conclusion of the test, the connectors and connector contacts, shall be washed, shaken, air blasted, and then permitted to dry for 24 hours at 40°C. The connectors shall be examined for destructive corrosion as defined in 3.20.1. The connector contacts shall be examined for destructive corrosion as defined in 3.20.2.

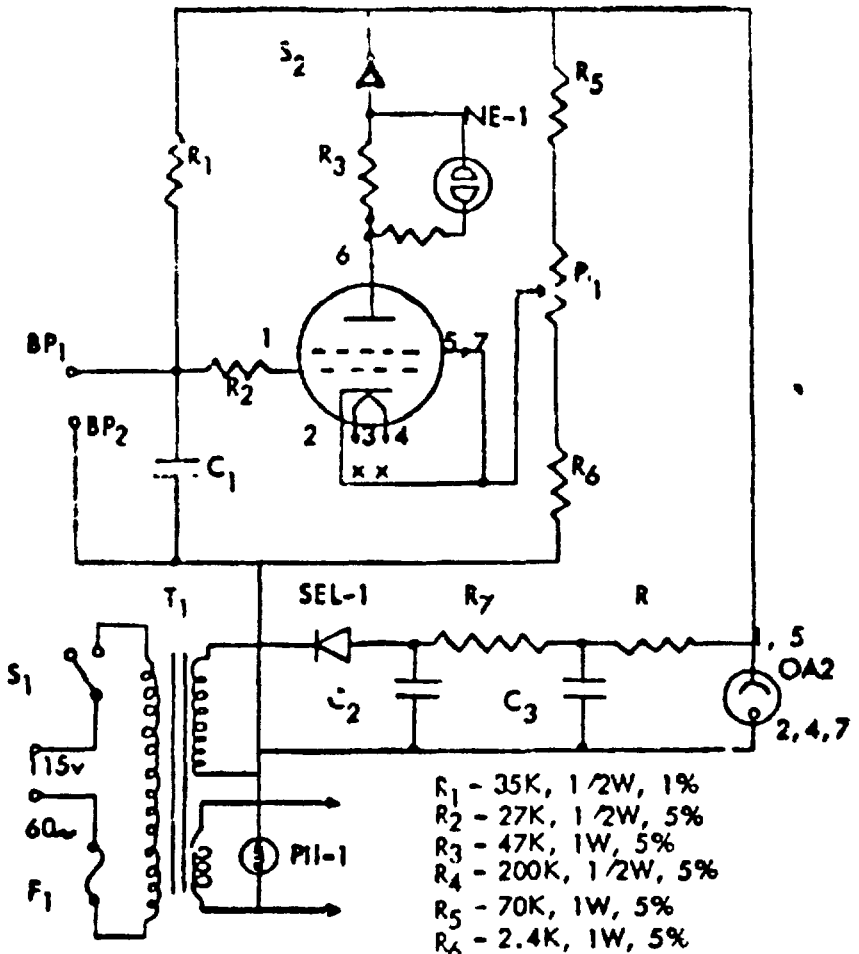
4.7.14 Moisture resistance.- Connectors mated and assembled to their appropriate cables shall be tested in accordance with Method 106A of Standard MIL-STD-202 except that the vibration sub-cycle shall not apply. (See 3.21)

4.7.15 Vibration.- Connectors mated and assembled to their appropriate cables, and adapters mated to a plug or jack, as applicable, assembled to its appropriate cable shall be tested in accordance with test condition A and C of Method 204A of Standard MIL-STD-202 except that the amplitude of the motion shall be a minimum of 20 gravity units. The connector assemblies shall be subjected to the specified cycles in each of the two mutually perpendicular directions. The test shall be conducted under the electrical load conditions of 4.7.16.1, or equivalent to monitor for interruptions of electrical contact of the mated connectors. (See 3.22 and 4.7)

4.7.16 Shock.- Connectors mated and assembled to their appropriate cables, and adapters mated to a plug or jack, as applicable, assembled to its appropriate cable, shall be tested in accordance with test condition d of Method 202A of Standard MIL-STD-202. The connector assemblies shall be subjected to six blows of 100 gravity units in each of the two perpendicular positions. The test shall be conducted under the electrical load conditions of Figure 1, or equivalent circuit which has 10 microsecond response to discontinuities to monitor for interruptions of electrical contact of the mated connectors. (See 3.22)

4.7.16.1 Operation of the monitor circuit.- The operation of the monitor circuit shall be as follows. (See Figure 1) - Values of $R_T C_1$, and the plate supply voltage shall be chosen so that the thyratron will fire when the duration of the discontinuity exceeds 10 microseconds. Restarting after an indication of failure shall be accomplished by operation of switch S2. The timing of the monitor circuit shall be adjusted by varying the positive voltage on the cathode of the thyratron through a plate supply to grid voltage divider. Potentiometer P_1 shall perform this function. Figure 2 shows a circuit for use in calibration. The plate of the twin triode (5AU7 (or equivalent) shall be connected to BP_1. The triode cathode shall be connected to BP_2 through a variable negative power supply. With both circuits operating and the thyratron non-conducting the voltage between BP_1, and BP_2 shall be adjusted to zero by variation of the triode cathode voltage. A pulse generator shall supply the triode grid with negative going pulses of at least -50 volts with a variable pulse duration of between 5 microseconds above and 5 microseconds below the specified value. Monitor circuit timing shall be viewed on an oscilloscope between BP_1, and BP_2. With microsecond timing markers on the oscilloscope trace, the pulse

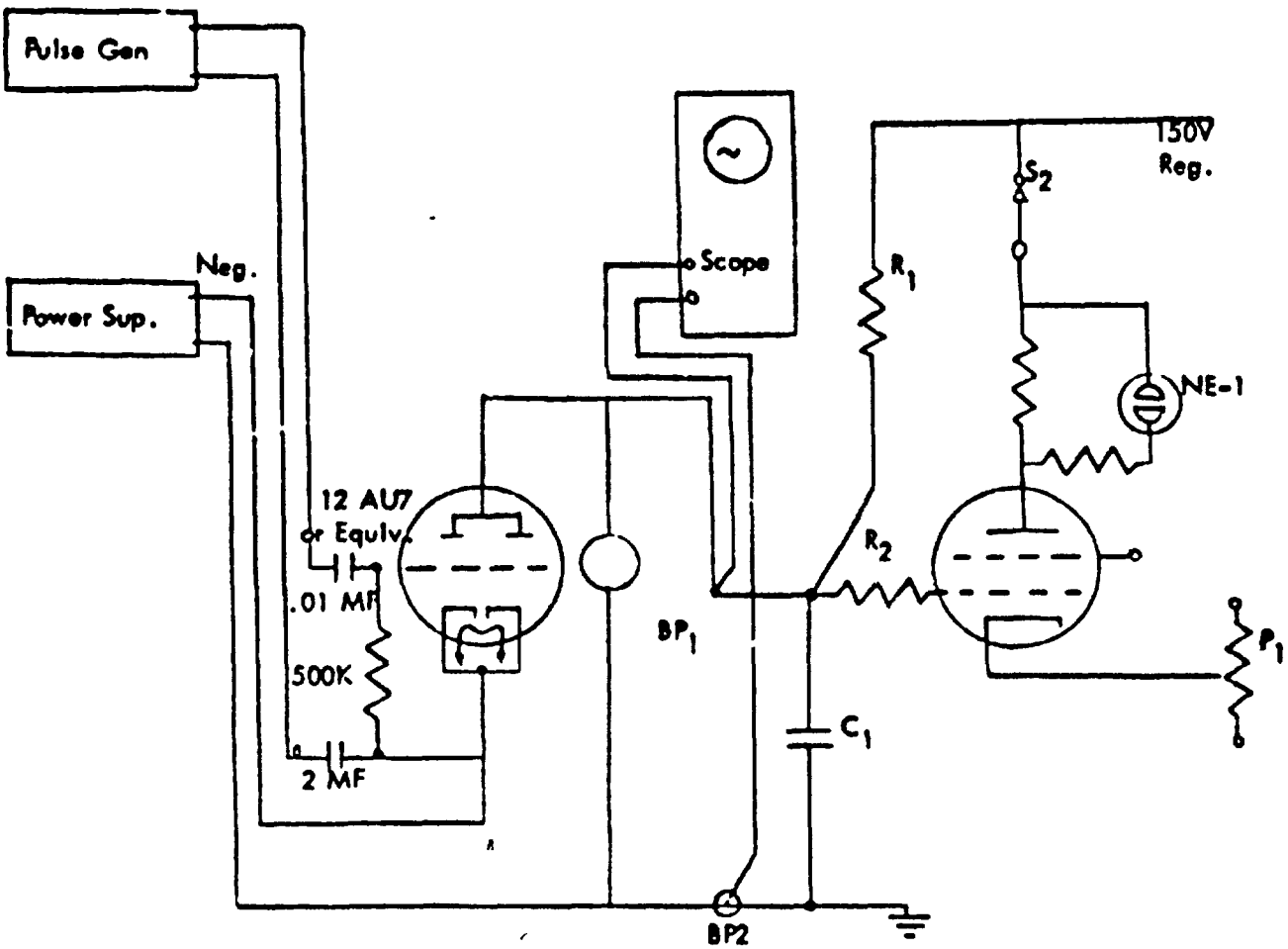
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R_1 - 35K, 1/2W, 1%
R_2 - 27K, 1/2W, 5%
R_3 - 47K, 1W, 5%
R_4 - 200K, 1/2W, 5%
R_5 - 70K, 1W, 5%
R_6 - 2.4K, 1W, 5%
R_7 - 1K, 1W, 5%
R_8 - 1K, 1W, 5%
C_1 - .0022 mfd, ±5% 600 DCWV
C_2 - 10 mfd, 400 DCWV
C_3 - 10 mfd, 400 DCWV
NE-1 NE-51
SEL-1-150v, 50 ma
S_1 - SPST, 125v, 2A (Toggle)
S_2 - SPST, (NC) 125v, 1A (Push)
Pil-1 - No. 47, 6.3v, Pilot Lamp
T_1 - PRI-117v, 60cps, 150v, 25ma, 6.3v
F_1 - Fuse 1a
P_1 - 5k - 1w
Tube 2D21 Thyratron

FIGURE 1 - Monitor for Electrical Discontinuity

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Pulse Gen: -30v min, pulse width 5/usec above to 5 usec below specified value.
Power Sup: 0 to -50 VDC Var.

FIGURE 2 - Calibration Circuit

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generator shall be adjusted to the pulse width as specified above. Potentiometer P_1 shall then be adjusted so that failure lamp NE-1 will just light. The setting of P_1 shall be checked by slowly increasing the pulse width from 5 microseconds below the specified value to assure lamp operation at precisely the specified value. This test shall cause points BP_1 and BP_2 to go from a zero voltage difference to essentially an infinite parallel impedance condition for a short period of time simulating the operation of the connectors. The interelectrode capacitance of the tube and the effects of wiring shall be considered negligible with respect to the impedance of the RC circuit.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Unless otherwise specified, connectors shall be individually protected and unit-packaged in accordance with Method 1A-8 of Specification MIL-P-116, without the use of contact preservatives. Prior to packaging, the parts of the connector shall be completely assembled. All center contacts, where loose, shall be securely fastened to the body of the connector by wire or other approved method. When applicable, a legible, concise set of assembly instructions shall be included with each cable-using connector. Unless otherwise specified five units packages or a multiple thereof shall be further packaged in intermediate containers conforming to Specification PPP-B-566, PPP-B-676, or MIL-B-4229. The gross weight of the intermediate container shall not exceed 10 pounds.

5.1.2 Level C.- Connectors shall be afforded preservation and packaging in accordance with the manufacturer's normal commercial practice.

5.2 Packing.-

5.2.1 Level A.- Connectors packaged as specified shall be packed in overseas-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (class 2 or 3, as specified) or wood-cleated paper-overlaid boxes conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, and MIL-B-10377, respectively, at the option of the manufacturer. Shipping containers shall have case liners conforming to Specification MIL-L-10547; the case liners shall be closed and sealed in accordance with the appendix thereto. Case liners boxes conforming to Specification PPP-B-636 may be omitted provided the center and edge seams and manufacturer's joints are sealed with tape, at least 1-1/2 inches wide, conforming to Type III, class 1, of Specification PPP-T-60. Box closures and strapping shall be as specified in the applicable box specification or appendix thereto. Fiberboard boxes shall be reinforced with bands of pressure sensitive filament tape conforming to Specification PPP-T-97 in lieu of metal strapping. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B.- Connectors packaged as specified shall be packed in domestic-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber, or wood-cleated paper-overlaid boxes conforming to Specifications PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, and MIL-B-10377, respectively, at the option of the manufacturer. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.3 Level C.- Connectors, packaged as specified, shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2.4 General.- Insofar as possible and practical, exterior containers shall be uniform in shape and size, shall be of minimum cube and tare consistent with the protection required, and shall contain identical quantities of identical items.

5.3 Marking.- In addition to any special marking required by the contract or order unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Connectors covered by this specification are intended for use in radio frequency applications up to 10,000 megacycles per second. They are designed for use with small size 50 ohm cables. Their use is governed by temperature limitations of their associated cables.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment (Level A, level B or level C).

(d) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(e) Place of final inspection.

6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

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6.4 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example UG-1354W/U. The contractor should apply for the nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.5 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Project Number 5935-A055

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